

Kyle Main

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Senior Security Data Scientist — Anomaly Detection, ML & Security Analytics

Security data scientist, 12 years, building statistical/ML detection systems against massive-scale telemetry — unsupervised clustering, time-series anomaly detection, and large-scale EDA (GCP Dataproc/PySpark) behind 220+ log sources and 2,300+ production detection rules across MITRE ATT&CK. Hands-on GenAI/LLM for security: prompt engineering for false-positive triage/rule generation, GenAI-orchestrated SIEM APIs. M.S. Physics (numerical data analysis & modeling).

DATA SCIENCE & MACHINE LEARNING

Languages & Libraries: Python, SQL, R; Pandas, scikit-learn, NumPy, SciPy, PyTorch

Modeling Techniques: Unsupervised/clustering ML (device behavior clustering on network telemetry), time-series anomaly detection (authentication volume/attempt anomalies, process-chain behaviors), statistical and behavioral detection modeling

Big Data & Cloud ML Platforms: GCP Dataproc/Zeppelin, PySpark/SparkSQL for large-scale EDA; Apache Beam/GCP Dataflow for high-volume batch retrieval; AWS, GCP, Azure cloud security

GenAI / LLM for Security: Prompt engineering for security use cases (false-positive triage, automated detection-content generation); GenAI-driven orchestration of SIEM APIs across customers/platforms; built reusable GenAI "skills" automating detection-rule translation between SIEM syntaxes

SECURITY DETECTION ENGINEERING & DATA PIPELINES

Detection Content: 2,300+ production detection rules (signature, statistical, behavioral, ML-based) across the MITRE ATT&CK matrix; multi-SIEM detection-as-code orchestration (Splunk, Sentinel, Defender, Google SecOps, CrowdStrike, SentinelOne) via native APIs, with GitLab CI/CD, automated testing, and staged rollout

Data Engineering: Pipelines ingesting 220+ unique log sources; 50+ Logstash normalization filters; Common Information Model (CIM) data-dictionary design; Elasticsearch (queries, transforms, Beats, detection rules, API), Splunk

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built an entirely new security data ingestion and analytics platform for DOE/NNSA from the ground up — CrowdStrike, Suricata, and Zeek telemetry into a central Elasticsearch environment, with a UEBA (User and Entity Behavior Analytics) detection layer built on custom data transforms to statistically surface anomalous entity behavior, plus data-quality monitoring/alerting and custom dashboards (DOE/NNSA Security Data Integration project, completed).
- Currently create and manage detection/alerting analytics directly supporting Treasury's Security Operations Center, translating security data patterns into actionable case content for incident response analysts (Treasury SOC / TSSOC, current project).
- Supported data ingestion/quality efforts in an Elasticsearch/Splunk environment for DOE's Continuous Diagnostics and Mitigation (CDM) program (completed).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building signature, statistical, behavioral, and ML-based detection models against massive-scale customer telemetry on a cloud-based big-data platform, translating analytical findings into production risk-detection content.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Built unsupervised clustering models to group network devices by behavioral feature similarity, and time-series anomaly-detection content for entity behaviors — authentication volume/attempt anomalies by country, and parent/child process-chain deviations — as core ML-based detection logic.
- Ran exploratory data analysis at scale on GCP Dataproc compute clusters using Zeppelin notebooks and a home-grown reusable analysis toolkit; wrote PySpark/SparkSQL jobs to load and analyze bucketed data, and a homegrown Apache Beam/GCP Dataflow program to retrieve high-volume historical cold-storage data for customer investigations.
- Created and managed 2,300+ individual detection rules covering most of the MITRE ATT&CK matrix as a very early hire, and owned data engineering for the pipeline feeding them — 220+ ingested log sources, 50+ Logstash normalization filters, and a Common Information Model standardizing fields across all parsed data.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security log data to develop custom statistical models for DNS-based malware detection/mitigation and anomalous network behavior discovery.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) — Numerical Data Analysis & Modeling, Applied Physics | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox

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August 3, 2026

Hiring Team

Amazon — AWS Security

I've spent 12 years building the statistical and machine-learning models that find cyber threats inside massive-scale infrastructure telemetry — exactly the anomaly detection, predictive modeling, and real-time risk-assessment work this AWS Security data scientist role is built around.

At Trend Micro/Cysiv, I built unsupervised clustering models to group network devices by behavioral feature similarity and time-series anomaly detection content covering authentication volume anomalies and process-chain deviations — production ML-based detection logic, not research exercises. I ran exploratory data analysis at scale on GCP Dataproc clusters using PySpark/SparkSQL and Zeppelin notebooks, and built a homegrown Apache Beam/Dataflow pipeline to retrieve high-volume historical data for investigations — the same big-data-processing muscle this role's SageMaker/EMR-scale work calls for, applied on a different hyperscale cloud. That statistical and ML detection work scaled into 2,300+ production detection rules across the MITRE ATT&CK matrix, built on a data-engineering pipeline I helped own spanning 220+ log sources.

More recently at DOE/NNSA, I built a UEBA detection layer directly on top of custom data transforms to statistically surface anomalous entity behavior — the same automated risk-assessment and pattern-recognition work described in this role's key responsibilities. I also bring direct, hands-on GenAI experience applied to security: prompt engineering for false-positive triage and automated detection-content generation, and using GenAI to orchestrate detection-rule deployment across SIEM APIs — real groundwork for expanding LLM agent pipelines in a security context. My M.S. in Physics centered on numerical data analysis and modeling, the quantitative foundation behind all of it.

I'd welcome the opportunity to bring that same statistical rigor and security-focused ML experience to protecting AWS customers at scale.

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